Employee Management Analysis - SQL

Create database

```
create database hr_db;
use hr_db;
```

Create tables

--Master tables

```
create table Departments(
	dept_id int primary key,
	dept_name varchar(50) not null
);

create table Locations(
	location_id int primary key,
	location_name varchar(50) not null
);

create table job_roles(
	role_id int primary key,
	role_name varchar(50) not null,
	min_salary decimal(10,2),
	max_salary decimal(10,2)
);
```

--Employee Table

```
create table Employees(
	emp_id int primary key,
	emp_name varchar(100) not null,
	gender varchar(10),
	dept_id int,
	role_id int,
	location_id int,
	join_date date,
	status varchar(20) default 'Active',
	foreign key (dept_id) references Departments(dept_id),
	foreign key (role_id) references job_roles(role_id),
	foreign key (location_id) references Locations(location_id)
);
```

--salary table

```
create table salaries(
	salary_id int primary key,
	emp_id int,
	salary decimal(10,2),
	effective_date date,
```

```
	foreign key(emp_id) references Employees(emp_id)
);
```

--performance table

```
create table performance_reviews(
	review_id int primary key,
	emp_id int,
	review_year int,
	rating decimal(2,1) check (rating between 1 and 5),
	foreign key(emp_id) references Employees(emp_id)
);
```

Inserting values

Departments table values

```
BULK INSERT Departments
FROM ‘C:/Users/Administrator/downloads/Departments.csv’
WITH(
	FIELDTERMINATOR=’,’,
	ROWTERMINATOR=’\n’,
	FRISTROW=2
);
```

Locations table values

```
BULK INSERT Locations
FROM ‘C:/Users/Administrator/downloads/Locations.csv’
WITH(
	FIELDTERMINATOR=’,’,
	ROWTERMINATOR=’\n’,
	FRISTROW=2
);
```

Job roles table values

```
BULK INSERT job_roles
FROM ‘C:/Users/Administrator/downloads/job_roles.csv’
WITH(
	FIELDTERMINATOR=’,’,
	ROWTERMINATOR=’\n’,
	FRISTROW=2
);
```

Employees table

```
BULK INSERT Employees
FROM ‘C:/Users/Administrator/downloads/Employees.csv’
WITH(
	FIELDTERMINATOR=’,’,
	ROWTERMINATOR=’\n’,
	FRISTROW=2
```

```
);
```

Salaries table

```
BULK INSERT salaries
FROM 'C:/Users/Administrator/downloads/salaries.csv'
WITH(
        FIELDTERMINATOR=',',
        ROWTERMINATOR='\n',
        FRISTROW=2
);
```

–performance table values

```
BULK INSERT performance_review
FROM 'C:/Users/Administrator/downloads/performance_review.csv'
WITH(
        FIELDTERMINATOR=',',
        ROWTERMINATOR='\n',
        FRISTROW=2
);
```

--Performance analysis

--1.Top 10 high performing employees

```
SELECT TOP 10 e.emp_id, e.emp_name, pr.rating
FROM Employees e
JOIN performance_reviews pr
ON e.emp_id = pr.emp_id
ORDER BY pr.rating DESC;
```

	emp_id	emp_name	rating
1	41	Vikram Sharma	5.0
2	58	Aditya Kumar	4.9
3	63	Neha Kumar	4.9
4	38	Ananya Reddy	4.9
5	82	Nisha Iyer	4.9
6	79	Arjun Kumar	4.8
7	62	Priya Reddy	4.8
8	57	Aditya Singh	4.8
9	40	Kiran Reddy	4.7
10	5	Sanjay Sharma	4.7

--2.Average rating per department

```
SELECT d.dept_name, AVG(pr.rating) AS avg_rating
FROM Employees e
JOIN Departments d ON e.dept_id = d.dept_id
JOIN performance_reviews pr ON e.emp_id = pr.emp_id
GROUP BY d.dept_name;
```

	dept_name	avg_rating
1	Customer Service	3.488888
2	Engineering	4.042857
3	Finance	3.794444
4	Human Resources	3.725000
5	IT Support	3.622222
6	Marketing	3.843750
7	Operations	3.550000
8	Sales	3.725000

--3.Employees eligible for promotion (rating >4 &2+years)

```
SELECT e.emp_id, e.emp_name, pr.rating
FROM Employees e
JOIN performance_reviews pr ON e.emp_id = pr.emp_id
WHERE pr.rating > 4
AND DATEDIFF(YEAR, e.join_date, GETDATE()) >= 2;
```

	emp_id	emp_name	rating
1	1	Pooja Sharma	4.6
2	5	Sanjay Sharma	4.7
3	10	Kiran Nair	4.5
4	11	Aditya Das	4.2
5	12	Ananya Reddy	4.3
6	19	Sneha Singh	4.7
7	22	Sneha Reddy	4.5
8	31	Rahul Singh	4.5
9	34	Pooja Nair	4.6
10	38	Ananya Reddy	4.9
11	40	Kiran Reddy	4.7
12	41	Vikram Sharma	5.0
13	43	Nisha Iyer	4.6
14	45	Divya Sharma	4.1
15	47	Rahul Sharma	4.6

	emp_id	emp_name	rating
16	48	Rahul Patel	4.4
17	49	Meera Iyer	4.5
18	51	Pooja Singh	4.2
19	56	Priya Singh	4.6
20	57	Aditya Singh	4.8
21	58	Aditya Kumar	4.9
22	59	Rahul Iyer	4.1
23	62	Priya Reddy	4.8
24	63	Neha Kumar	4.9
25	65	Pooja Nair	4.6
26	68	Arjun Das	4.3
27	79	Arjun Kumar	4.8
28	81	Sneha Nair	4.3
29	82	Nisha Iyer	4.9
30	84	Amit Patel	4.7

--4.performance trend over years

```
SELECT review_year, AVG(rating) AS avg_rating
FROM performance_reviews
GROUP BY review_year
ORDER BY review_year;
```

	review_year	avg_rating
1	2022	3.693750
2	2023	3.710344
3	2024	3.790000
4	2025	3.731578

--salary analysis

--5.salary distribution per department

```
select d.dept_name,
        MIN(s.salary) as min_salary,
        MAX(s.salary) as max_salary,
        AVG(s.salary) as avg_salary
from Employees e
```

```
join salaries s
on e.emp_id=s.emp_id
join Departments d
on e.dept_id=d.dept_id
group by d.dept_name;
```

	dept_name	min_salary	max_salary	avg_salary
1	Customer Service	297279.00	1236544.00	704589.555555
2	Engineering	317309.00	623662.00	474615.285714
3	Finance	271450.00	1396993.00	831651.444444
4	Human Resour...	459100.00	1239955.00	835522.187500
5	IT Support	526178.00	1242873.00	802562.611111
6	Marketing	284391.00	1258621.00	758580.125000
7	Operations	300544.00	1054663.00	672692.375000
8	Sales	324720.00	877157.00	578690.750000

--6.highest and lowest salary per role

```
select j.role_name,MIN(s.salary) as min_salary,
            MAX(s.salary) as max_salary
from Employees e
join salaries s
on e.emp_id=s.emp_id
join job_roles j
on e.role_id=j.role_id
group by j.role_name;
```

	role_name	min_salary	max_salary
1	Accountant	418170.00	706580.00
2	Customer Support Executive	271450.00	495567.00
3	HR Executive	317309.00	594169.00
4	Marketing Analyst	609634.00	791271.00
5	Operations Manager	772987.00	1279099...
6	Sales Manager	678404.00	1057180...
7	Software Engineer	636706.00	1396993...
8	System Administrator	502535.00	995068.00

--7.salary vs performance correlation

```
select e.emp_id,e.emp_name,s.salary,p.rating
from Employees e
join salaries s
on e.emp_id=s.emp_id
join performance_reviews p
on e.dept_id=p.emp_id;
```

	emp_id	emp_name	salary	rating
1	1	Pooja Sharma	576059.00	4.7
2	2	Rahul Das	1242900.00	4.7
3	3	Sanjay Kumar	1236544.00	2.9
4	4	Amit Sharma	896267.00	4.7
5	5	Sanjay Shar...	791271.00	4.6
6	6	Vikram Reddy	590744.00	2.9
7	7	Arjun Patel	849734.00	2.7
8	8	Kiran Das	300544.00	2.8
9	9	Kiran Sharma	858788.00	2.7
10	10	Kiran Nair	816538.00	4.6
11	11	Aditya Das	1000138.00	4.6
12	12	Ananya Re...	722973.00	2.9
13	13	Neha Singh	321690.00	2.9
14	14	Neha Sharma	1258621.00	4.7
15	15	Neha Sharma	515670.00	2.8

	emp_id	emp_name	salary	rating
16	16	Divya Patel	371209.00	3.9
17	17	Meera Iyer	1144341.00	4.0
18	18	Rohan Reddy	758898.00	2.9
19	19	Sneha Singh	769700.00	4.7
20	20	Neha Sharma	442985.00	4.0
21	21	Sanjay Reddy	1202010.00	4.7
22	22	Sneha Reddy	459100.00	4.6
23	23	Divya Kumar	931820.00	4.0
24	24	Sneha Kumar	706580.00	4.6
25	25	Sneha Shar...	1239955.00	4.6
26	26	Pooja Singh	931696.00	2.7
27	27	Amit Kumar	959950.00	4.6
28	28	Priya Sharma	384479.00	3.7
29	29	Rahul Reddy	713536.00	4.0
30	30	Rohan Das	804021.00	2.8

	emp_id	emp_name	salary	rating
31	31	Rahul Singh	967305.00	2.7
32	32	Divya Kumar	324720.00	3.9
33	33	Vikram Kumar	800755.00	2.7
34	34	Pooja Nair	552318.00	2.8
35	35	Vikram Das	622763.00	2.7
36	36	Rohan Reddy	763134.00	4.6
37	37	Meera Das	502282.00	3.9
38	38	Ananya Re...	772987.00	3.9
39	39	Priya Singh	594169.00	4.7
40	40	Kiran Reddy	688618.00	2.7
41	41	Vikram Shar...	622718.00	4.7
42	42	Arjun Das	736991.00	2.8
43	43	Nisha Iyer	851046.00	4.6
44	44	Kiran Iyer	652977.00	4.6
45	45	Divya Sharma	434275.00	2.8

	emp_id	emp_name	salary	rating
46	46	Nisha Nair	785725.00	4.0
47	47	Rahul Sharma	756128.00	4.0
48	48	Rahul Patel	636706.00	3.9
49	49	Meera Iyer	526178.00	2.7
50	50	Aditya Kumar	715833.00	2.7
51	51	Pooja Singh	495567.00	3.7
52	52	Divya Singh	726295.00	3.9
53	53	Aditya Reddy	317309.00	3.7
54	54	Ananya Das	725475.00	4.6
55	55	Nisha Sharma	986569.00	4.0
56	56	Priya Singh	609634.00	3.7
57	57	Aditya Singh	447587.00	4.7
58	58	Aditya Kumar	877157.00	3.9
59	59	Rahul Iyer	501481.00	2.9
60	60	Rahul Reddy	490119.00	4.0

	emp_id	emp_name	salary	rating
61	61	Rohan Kumar	288792.00	4.0
62	62	Priya Reddy	494853.00	3.7
63	63	Neha Kumar	271450.00	4.0
64	64	Arjun Iyer	608640.00	2.7
65	65	Pooja Nair	1279099.00	4.0
66	66	Amit Das	969123.00	4.7
67	67	Arjun Singh	895825.00	4.6
68	68	Arjun Das	870197.00	4.0
69	69	Sanjay Kumar	1054663.00	2.8
70	70	Vikram Nair	623662.00	3.7
71	71	Divya Das	802908.00	2.7
72	72	Aditya Kumar	916629.00	2.9
73	73	Sneha Das	1069351.00	4.0
74	74	Arjun Singh	914031.00	4.0
75	75	Vikram Das	1033030.00	4.6

	emp_id	emp_name	salary	rating
76	76	Amit Reddy	729120.00	4.6
77	77	Neha Nair	1082818.00	4.7
78	78	Kiran Kumar	1396993.00	4.0
79	79	Arjun Kumar	297279.00	2.9
80	80	Rohan Singh	1107602.00	4.0
81	81	Sneha Nair	284391.00	4.7
82	82	Nisha Iyer	502535.00	4.7
83	83	Priya Nair	842284.00	2.7
84	84	Amit Patel	1057180.00	4.0
85	85	Kiran Das	885959.00	2.7
86	86	Rahul Patel	396803.00	3.7
87	87	Rahul Patel	329138.00	4.7
88	88	Sanjay Iyer	680842.00	4.7
89	89	Rohan Nair	832952.00	2.7
90	90	Sneha Das	767115.00	4.6

90	90	Sneha Das	767115.00	4.6
91	91	Nisha Patel	463808.00	4.0
92	92	Ananya Iyer	764008.00	2.7
93	93	Meera Das	983057.00	2.8
94	94	Sneha Patel	418170.00	3.9
95	95	Sanjay Das	845292.00	2.7
96	96	Sneha Iyer	995068.00	2.9
97	97	Sanjay Reddy	977101.00	4.6
98	98	Vikram Kumar	1242873.00	2.7
99	99	Ananya Singh	678404.00	4.7
100	100	Pooja Das	659541.00	2.7

--8.Employees below average department salary

```
select e.emp_id,e.emp_name,s.salary
from Employees e
join salaries s
on e.emp_id=s.emp_id
where salary<(
                select AVG(s1.salary)
                from Employees e1
                join salaries s1
```

```
on e1.emp_id=s1.emp_id
where e1.dept_id=e.dept_id
);
```

	emp_id	emp_name	salary
1	5	Sanjay Sharma	791271.00
2	10	Kiran Nair	816538.00
3	22	Sneha Reddy	459100.00
4	24	Sneha Kumar	706580.00
5	36	Rohan Reddy	763134.00
6	44	Kiran Iyer	652977.00
7	54	Ananya Das	725475.00
8	76	Amit Reddy	729120.00
9	90	Sneha Das	767115.00
10	91	Nisha Patel	463808.00
11	60	Rahul Reddy	490119.00
12	61	Rohan Kumar	288792.00
13	63	Neha Kumar	271450.00
14	46	Nisha Nair	785725.00
15	47	Rahul Sharma	756128.00

	emp_id	emp_name	salary
16	29	Rahul Reddy	713536.00
17	20	Neha Sharma	442985.00
18	28	Priya Sharma	384479.00
19	86	Rahul Patel	396803.00
20	53	Aditya Reddy	317309.00
21	94	Sneha Patel	418170.00
22	37	Meera Das	502282.00
23	32	Divya Kumar	324720.00
24	16	Divya Patel	371209.00
25	1	Pooja Sharma	576059.00
26	39	Priya Singh	594169.00
27	41	Vikram Sharma	622718.00
28	57	Aditya Singh	447587.00
29	99	Ananya Singh	678404.00
30	87	Rahul Patel	329138.00

	emp_id	emp_name	salary
31	88	Sanjay Iyer	680842.00
32	81	Sneha Nair	284391.00
33	82	Nisha Iyer	502535.00
34	45	Divya Sharma	434275.00
35	8	Kiran Das	300544.00
36	15	Neha Sharma	515670.00
37	34	Pooja Nair	552318.00
38	35	Vikram Das	622763.00
39	33	Vikram Kumar	800755.00
40	40	Kiran Reddy	688618.00
41	49	Meera Iyer	526178.00
42	50	Aditya Kumar	715833.00
43	64	Arjun Iyer	608640.00
44	100	Pooja Das	659541.00
45	92	Ananya Iyer	764008.00

46	79	Arjun Kumar	297279.00
47	59	Rahul Iyer	501481.00
48	6	Vikram Reddy	590744.00
49	13	Neha Singh	321690.00

--Department growth

--9. Hiring trend per year

```
select year(join_date) as hire_date,
                COUNT(*) as total_hires
from Employees
group by YEAR(join_date)
order by hire_date;
```

	hire_date	total_hires
1	2018	22
2	2019	17
3	2020	21
4	2021	15
5	2022	17
6	2023	8

--10.headcount growth by department

```
select d.dept_name,COUNT(e.emp_id) as total_count
from Employees e
join Departments d
on e.dept_id=d.dept_id
group by d.dept_name;
```

	dept_name	total_count
1	Customer Service	9
2	Engineering	7
3	Finance	18
4	Human Resour...	16
5	IT Support	18
6	Marketing	16
7	Operations	8
8	Sales	8

--11.attrition rate by department

```
select d.dept_name,
                SUM(case when e.status='Inactive' then 1 else 0 end)as inactive_count,
                COUNT(*) as total_count
from Employees e
join Departments d
on e.dept_id=d.dept_id
group by d.dept_name;
```

	dept_name	inactive_count	total_count
1	Customer Service	2	9
2	Engineering	2	7
3	Finance	6	18
4	Human Resources	3	16
5	IT Support	4	18
6	Marketing	4	16
7	Operations	2	8
8	Sales	0	8

--strategic analysis

--12.Employees overdue for promotions

```
select e.emp_id,e.emp_name,e.join_date
from Employees e
where DATEDIFF(YEAR,e.join_date,GETDATE())>=7;
```

	emp_id	emp_name	join_date
1	2	Rahul Das	2019-11-04
2	4	Amit Sharma	2018-11-16
3	11	Aditya Das	2018-07-25
4	12	Ananya Reddy	2018-06-19
5	14	Neha Sharma	2018-01-17
6	16	Divya Patel	2018-03-21
7	21	Sanjay Reddy	2019-03-16
8	23	Divya Kumar	2018-02-08
9	25	Sneha Sharma	2018-11-29
10	26	Pooja Singh	2019-07-09
11	29	Rahul Reddy	2018-09-06
12	30	Rohan Das	2019-11-13
13	31	Rahul Singh	2019-08-30
14	34	Pooja Nair	2018-03-20
15	38	Ananya Reddy	2019-12-05

	emp_id	emp_name	join_date
16	39	Priya Singh	2019-09-13
17	41	Vikram Sharma	2018-04-21
18	44	Kiran Iyer	2019-05-05
19	46	Nisha Nair	2018-12-19
20	48	Rahul Patel	2019-09-21
21	55	Nisha Sharma	2018-09-26
22	60	Rahul Reddy	2018-03-11
23	62	Priya Reddy	2018-03-04
24	63	Neha Kumar	2018-06-07
25	64	Arjun Iyer	2019-04-16
26	67	Arjun Singh	2018-12-10
27	68	Arjun Das	2018-02-13
28	69	Sanjay Kumar	2019-09-11
29	71	Divya Das	2019-11-29
30	74	Arjun Singh	2018-11-02

31	76	Amit Reddy	2019-07-17
32	77	Neha Nair	2019-10-25
33	80	Rohan Singh	2018-02-22
34	82	Nisha Iyer	2018-03-16
35	89	Rohan Nair	2018-01-29
36	91	Nisha Patel	2019-04-11
37	93	Meera Das	2018-03-24
38	96	Sneha Iyer	2019-03-22
39	97	Sanjay Reddy	2019-06-25

--13.high salary but low performance cases

```
select e.emp_id,e.emp_name,s.salary,p.rating,p.review_year
from Employees e
join salaries s
on e.emp_id=s.emp_id
join performance_reviews p
on e.emp_id=p.emp_id
where s.salary>(
                select AVG(salary)
                from salaries)
and p.rating<3.0;
```

	emp_id	emp_name	salary	rating	review_year
1	7	Arjun Patel	849734.00	2.7	2023
2	18	Rohan Reddy	758898.00	2.8	2022
3	21	Sanjay Reddy	1202010.00	2.5	2025
4	33	Vikram Kumar	800755.00	2.9	2022
5	36	Rohan Reddy	763134.00	2.7	2022
6	46	Nisha Nair	785725.00	2.9	2025
7	66	Amit Das	969123.00	2.7	2022
8	72	Aditya Kumar	916629.00	2.9	2023
9	77	Neha Nair	1082818.00	2.6	2022
10	90	Sneha Das	767115.00	2.5	2024
11	93	Meera Das	983057.00	2.9	2022
12	96	Sneha Iyer	995068.00	2.7	2022

--16.Location wise salary comparison

```
select COUNT(e.emp_id) as total_employees,
        l.location_name,
        ROUND(avg(s.salary),2) as avg_salary,
        MAX(s.salary) as max_salary,
        MIN(s.salary) as min_salary
from Employees e
join salaries s
on e.emp_id=s.emp_id
join Locations l
on e.location_id=l.location_id
group by l.location_name
order by avg_salary desc;
```

	total_employees	location_name	avg_salary	max_salary	min_salary
1	17	Bangalore	816903.590000	1396993.00	288792.00
2	20	Pune	789646.550000	1242873.00	329138.00
3	28	Mumbai	733655.430000	1258621.00	297279.00
4	19	Chennai	715993.210000	1082818.00	384479.00
5	16	Hyderabad	673103.310000	1279099.00	271450.00

--17.gender diversity ratio

```
select gender,
            COUNT(*) as total_employees,
            ROUND(count(*)*100.0/(select count(*) from Employees),2) as percentage
from Employees
group by gender;
```

	gender	total_employees	percentage
1	Female	52	52.000000000000
2	Male	48	48.000000000000

--18.median salary calculation

```
SELECT DISTINCT
PERCENTILE_CONT(0.5)
WITHIN GROUP (ORDER BY salary)
OVER () AS median_salary
FROM salaries;
```

	median_salary
1	757513